

Appendix I – SGIF History

Rural Fire Information Management System - SGIF

In 2001, a protocol was established between the National Center for Geographic Information (CNIG) and the Directorate-General for Forests (DGF) aimed at producing a computer application designated as the Forest Fire Information Management System (SGIF), with the objective of managing the national database on rural fires.

In the same year, an application was installed in all Prevention and Detection Centers (CPDs), which allowed the computerization of the rural fire data registration process in a systematic and uniform manner and took the first steps towards digital data transmission to a single central database.

The data structure and application development were established in early 2001, with installation and operator training taking place at the beginning of the fire season in June.

From this outset, the system underwent several evolutions as needs arose, driven by technological advancements and institutional reorganizations in the forestry and civil protection sectors.

The following summary presents significant dates in the history of SGIF.

<i>Year</i>	<i>Actions</i>
2001	Protocol established between the National Center for Geographic Information (CNIG) and the Directorate-General for Forests (DGF) for the creation of SGIF. Development of the application and central database. Acquisition of computers and modems. Installation of SGIF application in CPDs.
2002	Conversion of the central database to SQL Server. Opening to import data directly from the National Fire Service (SNB) application.
2003	Installation of permanent internet in CPDs. Improvement in the process of importing data to the central database.
2006	Conversion of the application to a web application. Registration operation is now carried out by the National Republican Guard (GNR), and CPDs are extinguished, and Forest Information Maintenance and Exploitation Teams (EMEIF) are created. Records are now created by the District Command of Relief Operations (CDOS) and imported directly to the central database. Study conducted by COTEC Portugal on surveillance using SGIF data.
2009	Development of a new version. SGIF starts to have meteorological data associated with fires. SGIF starts to control access to registration and entry into the system. Modules developed for Forestry Technical Offices (GTF). Development of a statistical module for public access.
2011	Initiation of trend studies.
2012	Improvement of data consistency verification modules. The location of occurrences is now done by the National Authority for Emergency and Civil Protection (ANEPC). Adjustment of the rekindling concept.
2013	Update of the new list of parishes. Prototype created for authorization of burns and controlled burns for public access.
2014	Modules developed for the management of Municipal Forest Defense plans and awareness. A new module is developed for the management of controlled fire activity.
2015	Current version of SGIF produced. Development of a statistical analysis portal initiated.

2016	Revision of the data inconsistency evaluation module. Critical site analysis is instituted, with the use of Heat Maps for the distribution of occurrences.
2018	Areas burned over 10 hectares are digitized by an ICNF team, with the filling of parcel areas done automatically by the perimeter of the area. A new verification is instituted to check if the starting point is inside or outside the burned area perimeter. A new version of the burning and controlled burning registration system is made available to the public. The geographical component of burned area perimeters is stored in an SQL server database.
2021	The meteorological data series now only has one data source, using the ERA5 global reanalysis series from the European Centre for Medium-Range Weather Forecasts (ECMWF).
2022	The encoding of occurrences becomes the same in the SADO system of ANEPC and SGIF, adopting the SADO encoding. Coding lists for causes and alert lists for occurrences are revised. Various geographic data services with SGIF information are made available.
2024	The meteorological data series was extended until 1980, using the ERA5 global reanalysis series from the European Centre for Medium-Range Weather Forecasts (ECMWF).

Important Milestones

At the end of 2001, the process began that led to the conversion of the central database to a SQL Server database, which is still in use today. In 2006, it was necessary to develop a new version due to the changes in responsibilities resulting from the application of concepts established in the National Forest Fire Defense Plan. In 2009, a public access statistical module was launched. In 2018, there was a much greater integration between alphanumeric data and geographic data, and the first public version of the permit to burn registration system was launched.

Where the SGIF Project Fits In

SGIF was part of a broader project, initiated in 1999 at the National Geographic Information Center, called the Emergency Situations Information Network (RISE). The main objectives of the Emergency Situations Information Network were to equip the country with an electronic network of georeferenced information, providing an automatic and up-to-date flow of data related to emergency situations. A subset of this data was made available to the public through the National Geographic Information System website. RISE was a service developed by the Forest Fire Risk Cartography Group (CRIF) of the CNIG, in close cooperation with the National Fire Service, the National Civil Protection Service, the Directorate-General for Forests, the Geographic Institute of the Army, and, at the time of the CNIG's dissolution, agreements were being established with the Directorate-General for Traffic and the National Institute of Medical Emergency. Through a protocol, the adherence of entities that generated records related to emergency situations to the RISE was formalized, ensuring:

- Remote uploading of information to the database of that institution or to the database of another institution adhering to this protocol.

- Consultation of cartographic information produced by the Geographic Institute of the Army, made available by RISE applications.
- The use of information provided by RISE requires institutions to adhere to a set of conditions, including the exclusive use of cartographic data strictly within the scope of emergency or for internal purposes, prohibiting its reproduction or transfer to third parties.
- This cooperation with other institutions in the network resulted in benefits in terms of sharing experiences, reducing costs, and optimizing benefits.
- There was an exchange of scientific and technical information related to the management of georeferenced emergency information.
- There was a strengthening of national intervention capacity in the development of international cooperation projects, which was recognized, for example, by the European Space Agency. In 2001, this agency opened a specific application process to support the development of this network, which resulted in the "PremFIRE" project (CNIG, 2001)

What is the SGIF today and how Is It maintained?

Today, SGIF is a system that collects information on rural fires and generates statistical data on the same topic. It is the national reference database for rural fires. After the creation of the SGIF in 2001, the first major organizational change occurred in 2006. The SGIF became the national reference database for rural fires, which required it to adapt, evolving from an information management application of the DGF to a system that integrates information from the three key institutions of Forest Fire Defense (DFCI) (National Civil Protection Authority - ANPC, National Republican Guard - GNR, and National Forestry Authority (AFN). The second major organizational change took place after 2017, with the introduction of the Integrated Rural Fire Management System. Despite the profound changes during this period, no significant alterations were required in the operational structure of the SGIF. Currently, the SGIF has about 700 users, and the information it produces is used by the Institute for the Conservation of Nature and Forests (ICNF), the National Republican Guard, and the National Emergency and Civil Protection Authority (ANEPC) to produce numerous official reports. Additionally, various scientific studies and these have utilized SGIF data. The responsibility for maintaining the SGIF lies with the ICNF, but since the dissolution of the CNIG in 2002, all developments were carried out by Rui Almeida, João Moreira, Raquel Onofre and Pedro Venâncio.

SGIF has undergone several improvements over time. In 2010, a thorough analysis of the data was conducted. This process required an exhaustive review of the entire rural fire database, during which some inconsistencies were detected, necessitating the republication of statistical series and base data.

Since 2010, the system has evolved to include stricter internal procedures for data consistency verification. Procedures for associating other information, such as meteorological data (observed and forecasted values), have also been implemented.

The data were re-evaluated again in 2015 and 2022, with the decision to use a new meteorological data series to ensure uniformity in the data source. Whenever it has been

possible to associate new data with the core rural fire information, it has enabled the development of new studies, such as the one in this thesis.

One of the current objectives of SGIF is to expand the set of users accessing its information. To this end, new modules for unregistered users are being developed. Examples include partnerships being developed with Pordata and the National Institute of Statistics, where new approaches to disseminating information are being created.

Appendix II – Metadata on SGIF Information

The SGIF SQL database contains systematically organized rural fire data since 2001. It also includes data on other topics related to rural fires, such as meteorology, orography, infrastructure, and land use. The following table presents the list and description of the variables associated with each rural fire.

VARIABLE	DESCRIPTION
CODIGO	SGIF code for the fire – Forest Service code
CODIGOSADO	SADO code for the fire – Civil protection Code
ANO	Year of the alert date
TIPO	Forestal or Agricultural
DHINICIO	Date/Time of the alert
DH1INTERVENCAO	Date/Time of the first intervention
DHRESOLUCAO	Date/Time of the resolution
DHCONCLUSAO	Date/Time of the conclusion
DHFIM	Date/Time of extinction
DURACAOHORAS	Duration in number of hours
AREAPOV	Burned area of forest stands (ha)
AREAMATO	Burned area of shrublands (ha)
AREAAGRIC	Burned area of agricultural lands (ha)
AREATOTAL	Total burned area (ha)
NUTS2	NUTS II region of the fire's point of origin
NUTS3	NUTS III region of the fire's point of origin
DISTRITO	District of the fire's point of origin
CONCELHO	Municipality of the fire's point of origin
FREGUESIA	Parish of the fire's point of origin
LOCAL	Nearest location to the fire's point of origin
INE	INE code of the parish at the fire's point of origin
PERIMETRO	National forest or forest perimeter at the point of origin
APS	Protected area (RNAP) at the point of origin
X_20790	X coordinate in EPSG:20790 Lisbon (Lisbon) / Portuguese National Grid
Y_20790	Y coordinate in EPSG:20790 Lisbon (Lisbon) / Portuguese National Grid
LAT_4326	Latitude in EPSG:4326 WGS 84 -- WGS84 - World Geodetic System 1984
LON_4326	Longitude in EPSG:4326 WGS 84 -- WGS84 - World Geodetic System 1984
X_3763	X coordinate in EPSG:3763 ETRS89 / Portugal TM06
Y_3763	Y coordinate in ETRS89 (PT-TM06)
MODFARSITE	Farsite model of the burn area at the point of origin
AREAMANCHAMODFARSITE	Burn area of the Farsite model at the point of origin (ha)
ALTITUDEMEDIA	Average altitude of the 1x1 km grid at the point of origin
DECLIVEMEDIO	Average slope of the 1x1 km grid at the point of origin (%)
DENSIDADERV	Road density in the 1x1 km grid (m/ha)
COSN5VARIEDADE	Number of different COS N5 classes in the 1x1 km grid
PERIGOSIDADE	Structural hazard

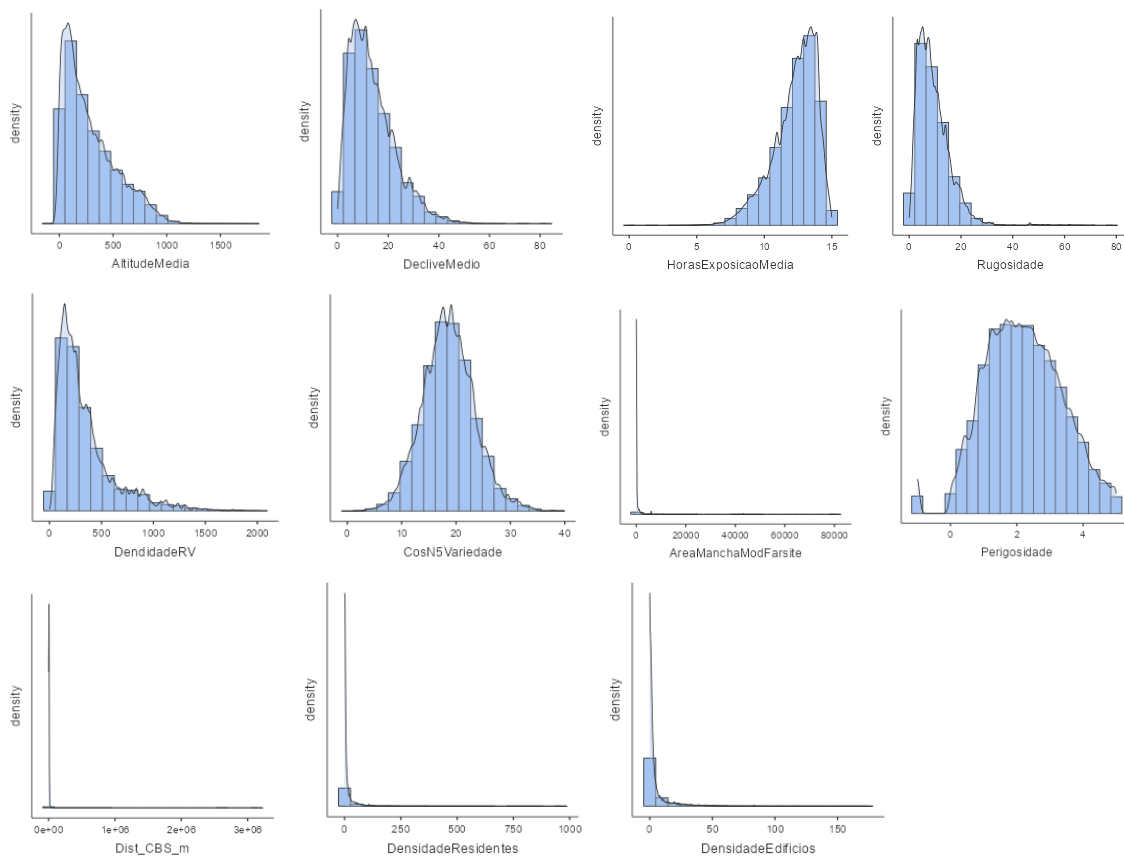
FWI	Meteorological index
DSR	Meteorological index
ISI	Meteorological index
DC	Meteorological index
DMC	Meteorological index
FFMC	Meteorological index
BUI	Meteorological index
THC	Meteorological index
DIST_CBS_M	Straight-line distance to the nearest fire station (m)
CBS	Nearest fire station in a straight line
NINCSIMULDISTRITO	Number of fires that were simultaneously active (fully or partially) with the recorded fire and had their point of origin in the same district as the recorded fire
NINCSIMULCONCELHO	Number of fires that were simultaneously active (fully or partially) with the recorded fire and had their point of origin in the same municipality as the recorded fire
NINCSIMULDISTRITO90	Number of fires that were simultaneously active (fully or partially) during the first 90 minutes of the recorded fire (after the alert date/time) and had their point of origin in the same district as the recorded fire
NINCSIMULCONCELHO90	Number of fires that were simultaneously active (fully or partially) during the first 90 minutes of the recorded fire (after the alert date/time) and had their point of origin in the same municipality as the recorded fire
NINCSIMUL500090	Number of fires that were simultaneously active (fully or partially) during the first 90 minutes of the recorded fire (after the alert date/time) and had their point of origin within a 5000-meter radius around the recorded fire's point of origin
DISTINCSIMUL500090	Number of fires that were simultaneously active (fully or partially) during the first 90 minutes of the recorded fire (after the alert date/time) and had their point of origin in the same district and within a 5000-meter radius around the recorded fire's point of origin
AREATOTALINCSIMUL500090	Sum of the burned areas of all fires that were simultaneously active (fully or partially) during the first 90 minutes of the recorded fire (after the alert date/time) and had their point of origin in the same district and within a 5000-meter radius around the recorded fire's point of origin
CAUSA	Numerical code for the fire's cause
TIPOCAUSA	Type of cause
GRUPOCAUSA	Cause group
DESCRICAO	Description of the cause
REACENDIMENTO_INCENDIOPAI	If reignited, code of the parent fire
ORIGINOUREACENDIMENTO	If it caused reignition (0 or 1)
FONTEALERTA	Source of the alert
CLASSEAREA	Burned area class
QO	Operational grid identification (1 x 1 km)
TEMPO1INTERVENCAO	Time between the alert and the first intervention (minutes)
TEMPORESOLUCAO	Time between the first intervention and resolution (minutes)
TEMPORESCALDO	Time between resolution and conclusion (minutes)

VELOCIDADEPROPAGACAO	Average rate of fire spread (ha/hour)
RCM	RCM or PIR
TEMPERATURA	Temperature
HUMIDADERELATIVA	Relative Humidity
VENTOINTENSIDADE	Wind intensity (Km/h)
VENTOINTENSIDADE_VETOR	Wind intensity vector (Km/h)
VENTODIRECAO_VETOR	Wind direction vector (Degrees)
PRECIPITACAO	Precipitation
VENTODIRECAO	Wind direction (Degrees)
HFFMC	FFMC index at the time of the alert
HISI	ISI index at the time of the alert
HFWI	FWI index at the time of the alert
MAXFWIH_48H_POSEXTINCAO	Maximum FWI index in the 48 hours after extinction
MAXFFMCH_48H_POSEXTINCAO	Maximum FFMCH index in the 48 hours after extinction
MAXISIH_48H_POSEXTINCAO	Maximum ISI index in the 48 hours after extinction
MAXDC_48H_DIAPOSEXTINCAO	Maximum DC index in the 48 hours after extinction
MAXBUI_48H_POSEXTINCAO	Maximum BUI index in the 48 hours after extinction
MAXDMC_48H_POSEXTINCAO	Maximum DMC index in the 48 hours after extinction

Appendix III – New Data Fields and Descriptive Statistical Overview

Based on the initial classification of new fields we present a descriptive analysis of the first two groups social and physical data.

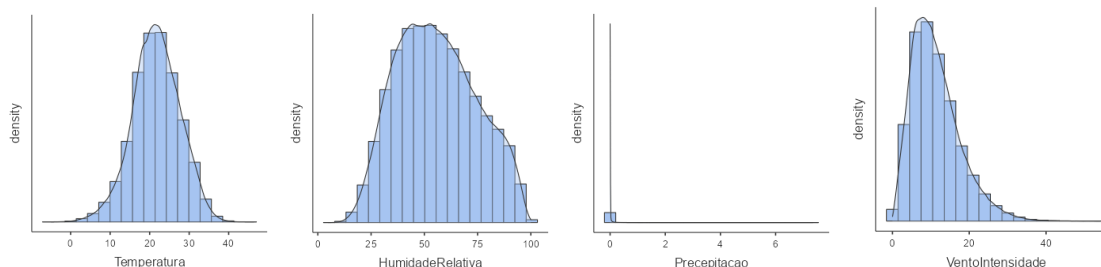
	N	Missing	Mean	SE	Median	SD	Variance	Minimum	Maximum	Skewness		Kurtosis	
										Skewness	SE	Kurtosis	SE
AltitudeMedia	809826	165	284.61	0.26771	217.24	240.92	58040.82	-64.21	1848.80	0.9715	0.00272	0.330	0.00544
DecliveMedio	809826	165	13.69	0.00979	11.83	8.81	77.62	0.00	82.34	1.0091	0.00272	0.996	0.00544
HorasExposicaoMedia	809967	24	12.28	0.00176	12.59	1.59	2.52	0.00	15.00	-0.9359	0.00272	0.724	0.00544
Rugosidade	809967	24	9.30	0.00686	8.05	6.18	38.15	0.00	78.28	1.9285	0.00272	9.637	0.00544
DendidadeRV	809849	142	338.92	0.30194	255.87	271.72	73833.72	0.00	2041.13	1.7721	0.00272	3.601	0.00544
CosN5Variedade	809858	133	18.62	0.00526	18.55	4.73	22.39	1.00	39.88	0.1546	0.00272	0.275	0.00544
AreaManchaModFarsite	198809	611182	637.98	8.29618	28.79	3699.10	1.37e+7	1.00	80409.60	13.5592	0.00549	229.482	0.01099
Perigosidade	809991	0	2.21	0.00127	2.16	1.15	1.31	-1.00	5.00	0.0877	0.00272	-0.303	0.00544
Dist_CBS_m	809991	0	4374.19	6.40907	3805	5768.13	3.33e+7	0	3141387	316.3321	0.00272	157746.284	0.00544
DensidadeResidentes	796826	13165	16.13	0.04518	3.62	40.33	1626.71	0.00	961.31	5.6504	0.00274	42.730	0.00549
DensidadeEdificios	799338	10653	5.47	0.01292	1.63	11.56	133.52	0.00	173.41	4.9115	0.00274	35.259	0.00548

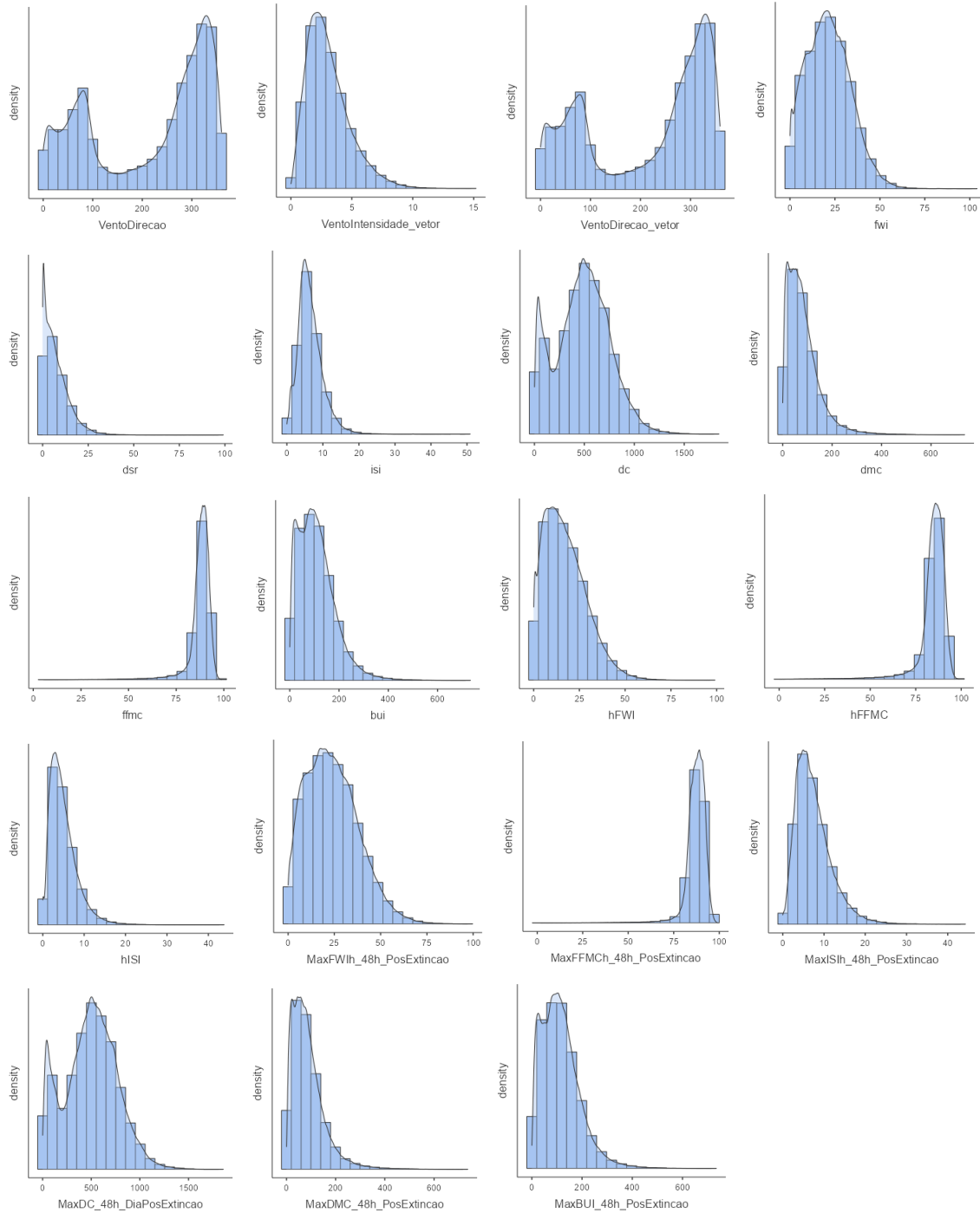


The data shows that for altitude (AltitudeMedia), with a mean of 284.61m and a standard deviation of 240.92, the slight positive skewness and low kurtosis suggest the median is more appropriate to use due to a tail towards higher values. Similarly, for slope (DecliveMedio), the mean slope is 13.69% with a standard deviation of 8.81, but the positive skewness and near-normal kurtosis indicate the median is a better representation. Exposure hours (HorasExposicaoMedia) have a mean of 12.28 hours and a standard deviation of 1.59, with

negative skewness and slight kurtosis, suggesting the median is better due to the tail towards lower exposure hours. Terrain rugosity (Rugosidade), with a mean of 9.30 and a standard deviation of 6.18, shows high positive skewness and kurtosis, making the median more suitable due to the presence of outliers. For road density (DensidadeRV), with a mean of 338.92 and a standard deviation of 271.72, the high skewness and moderate kurtosis also favor using the median. In contrast, variability on land use (CosN5Variedade), with a mean of 18.62 and low skewness and kurtosis, is better represented by the mean. However, the size of fuel model area (AreaManchaModFarsite), which has a mean of 637.98 units and a very high standard deviation of 3699.10, along with extreme skewness and kurtosis, is better represented by the median due to significant outliers. The danger index (Perigosidade) has a mean of 2.21 and low skewness and kurtosis, making the mean suitable. Distance to fire brigades (Dist_CBS_m), with a mean of 4374.19 meters and a very high standard deviation of 5768.13, shows extreme skewness and kurtosis, indicating that the median is more appropriate. Lastly, for social information both density of residents (DensidadeResidentes) (mean of 16.13) and building density (DensidadeEdificios) (mean of 5.47) have high skewness and kurtosis, suggesting that the median is better due to the presence of higher values and outliers.

	N	Missing	Mean	SE	Median	SD	Variance	Minimum	Maximum	Skewness		Kurtosis	
										Skewness	SE	Kurtosis	SE
Temperatura	806296	3695	21.6547	0.00671	21.65	6.028	36.3419	-4.93409	46.37	-0.140	0.00273	0.0949	0.00546
HumidadeRelativa	806296	3695	55.8400	0.02074	54.62	18.621	346.7281	4.83346	100.01	0.186	0.00273	-0.7654	0.00546
Precepitacao	806296	3695	0.0142	1.34e-4	0.00	0.120	0.0144	-8.67e-16	7.35	19.603	0.00273	572.0749	0.00546
VentoIntensidade	806296	3695	11.3318	0.00703	10.27	6.312	39.8448	0.00414	54.08	0.975	0.00273	1.2390	0.00546
VentoDirecao	806287	3704	215.5281	0.13238	268.95	118.866	14129.1731	2.06e-4	360.00	-0.489	0.00273	-1.3810	0.00546
VentoIntensidade_vetor	806287	3704	3.0590	0.00194	2.74	1.746	3.0489	0.00621	14.79	1.031	0.00273	1.3512	0.00546
VentoDirecao_vetor	806287	3704	215.6805	0.13349	270	119.865	14367.5488	0	359	-0.505	0.00273	-1.3788	0.00546
fwi	806287	3704	21.5435	0.01340	20.90	12.031	144.7464	3.15e-8	101.26	0.417	0.00273	-0.0205	0.00546
dsr	806287	3704	7.5585	0.00755	5.90	6.779	45.9603	1.44e-15	96.42	1.590	0.00273	4.7763	0.00546
isi	806287	3704	6.4578	0.00376	5.95	3.373	11.3781	1.55e-7	49.51	1.156	0.00273	3.6434	0.00546
dc	806287	3704	485.0279	0.29433	492.32	264.292	69850.3065	0.32922	1798.09	0.138	0.00273	-0.2948	0.00546
dmc	806287	3704	82.1325	0.07212	68.62	64.762	4194.1551	2.77e-5	716.23	1.854	0.00273	5.9668	0.00546
ffmc	806287	3704	87.5541	0.00694	88.62	6.236	38.8838	5.32149	98.92	-3.890	0.00273	24.2077	0.00546
bui	806287	3704	109.0479	0.08247	98.80	74.050	5483.4111	0.00000	713.88	1.155	0.00273	2.4007	0.00546
hFWI	806287	3704	17.3415	0.01272	15.63	11.418	130.3623	6.38e-10	96.37	0.797	0.00273	0.5513	0.00546
hFFMC	806287	3704	84.4498	0.00889	85.76	7.985	63.7600	0.01848	98.93	-3.759	0.00273	22.1789	0.00546
hISI	806287	3704	4.9238	0.00358	4.25	3.216	10.3440	2.86e-9	42.63	1.443	0.00273	3.6495	0.00546
MaxFWIh_48h_PosExtincao	805719	4272	23.9518	0.01553	22.66	13.942	194.3889	4.64e-9	97.00	0.554	0.00273	0.0284	0.00546
MaxFFMCh_48h_PosExtincao	805719	4272	87.2355	0.00638	87.90	5.725	32.7780	1.89429	99.18	-3.749	0.00273	29.3861	0.00546
MaxISih_48h_PosExtincao	805719	4272	7.4215	0.00476	6.62	4.272	18.2522	2.13e-8	43.11	1.211	0.00273	2.4020	0.00546
MaxDC_48h_DiaPosExtincao	805719	4272	497.5105	0.29603	506.64	265.723	70608.7166	0.32922	1807.73	0.110	0.00273	-0.2969	0.00546
MaxDMC_48h_PosExtincao	805719	4272	86.8572	0.07331	73.67	65.808	4330.7507	0.04299	717.36	1.802	0.00273	5.6903	0.00546
MaxBUI_48h_PosExtincao	805719	4272	114.5755	0.08341	105.04	74.872	5605.8229	0.06482	714.88	1.112	0.00273	2.2835	0.00546



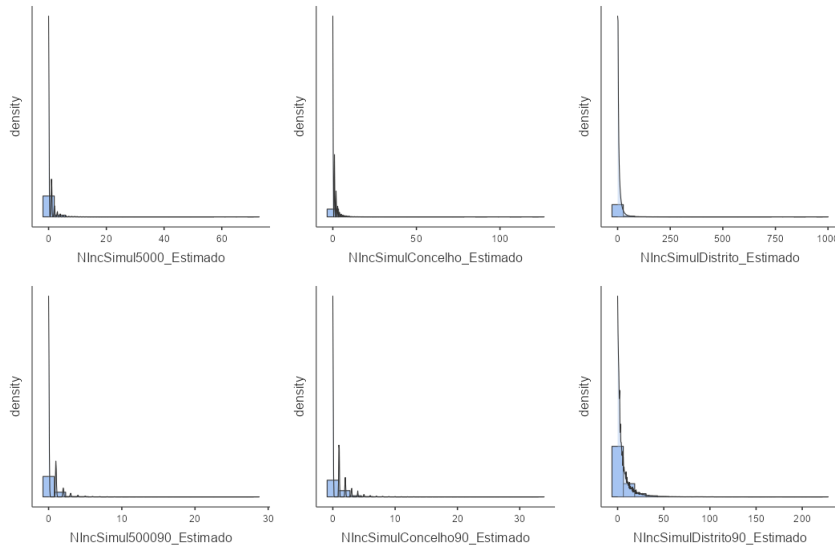


The data on meteorological group reveals that for temperature (Temperatura), with a mean of 21.65°C and a standard deviation of 6.03, the slight negative skewness and low kurtosis suggest the median is appropriate due to the slight tail towards lower temperatures. For relative humidity (HumidadeRelativa), the mean is 55.84% with a standard deviation of 18.62. The positive skewness and negative kurtosis indicate that the mean is a reliable measure, though the median could also be considered due to the tail towards higher humidity values. Precipitation (Precipitação) has a mean of 0.0142 mm and a very low

standard deviation of 0.12, but with extreme positive skewness and kurtosis, indicating the presence of outliers. Hence, the median is a more suitable representation. For wind intensity (VentoIntensidade), the mean wind speed is 11.33 km/h with a standard deviation of 6.31, and moderate positive skewness along with slight leptokurtosis suggests the median is more appropriate due to a tail towards higher wind speeds. Wind direction (VentoDirecao) and wind direction resultant for all time of fire (VentoDirecao_vetor) are not adequate to be analysed as linear continuous variable despite showing a slight negative skewness and negative kurtosis, indicating the data is slightly left-skewed and platykurtic. Fire Weather Index (FWI), with a mean of 21.54 and a standard deviation of 12.03, shows slight positive skewness and near-normal kurtosis, indicating the median might be slightly better due to a rightward tail. DSR (Daily Severity Rating) has a mean of 7.56 and a standard deviation of 6.78, with significant positive skewness and high kurtosis, suggesting the median is more appropriate due to extreme values. For Initial Spread Index (ISI), the mean is 6.46 with a standard deviation of 3.37, and both high positive skewness and kurtosis indicate that the median is better due to the presence of outliers. Drought Code (DC) and Duff Moisture Code (DMC) have means of 485.03 and 82.13, respectively, with standard deviations of 264.29 and 64.76. Both show positive skewness and varying kurtosis, suggesting the median is better for representing central tendencies, particularly for DMC with its extreme skewness and kurtosis. For Fine Fuel Moisture Code (FFMC), the mean is 87.55 with a standard deviation of 6.24, but very high negative skewness and extreme kurtosis indicate that the median is more suitable due to significant lower-end outliers. Build-Up Index (BUI), with a mean of 109.05 and a standard deviation of 74.05, has high positive skewness and moderate kurtosis, suggesting the median is better due to a tail towards higher values. Hourly FWI (hFWI) and hourly ISI (hISI) also show high skewness and kurtosis, making the median more appropriate for these variables. Conversely, maximum of hourly FWI in the 48 hours pos fire (MaxFWIh_48h_PosExtincao) and maximum of hourly FWI in the 48 hours pos fire (MaxISih_48h_PosExtincao) have skewness and kurtosis values that indicate a median would better represent the data due to extreme outliers.

Overall, for most variables with high skewness and kurtosis, the median is a more reliable measure of central tendency, especially where significant outliers or non-normal distributions are present.

	N	Missing	Mean	SE	Median	SD	Variance	Minimum	Maximum	Skewness		Kurtosis	
										Skewness	SE	Kurtosis	SE
NIncSimul5000_Estimado	809991	0	0.477	0.00165	0	1.488	2.213	0	71	9.01	0.00272	141.5	0.00544
NIncSimulConcelho_Estimado	803721	6270	0.921	0.00284	0	2.548	6.493	0	123	8.77	0.00273	132.4	0.00546
NIncSimulDistrito_Estimado	803721	6270	8.693	0.02485	3	22.278	496.311	0	974	9.14	0.00273	127.8	0.00546
NIncSimul500090_Estimado	809991	0	0.351	0.00108	0	0.972	0.945	0	28	6.35	0.00272	71.8	0.00544
NIncSimulConcelho90_Estimado	805110	4881	0.670	0.00177	0.00	1.589	2.525	0	33	6.22	0.00273	65.1	0.00546
NIncSimulDistrito90_Estimado	805110	4881	6.227	0.01400	2.00	12.560	157.764	0	222	6.95	0.00273	72.9	0.00546



The data resulting from reanalysis for number of simultaneous fires in radius of 5000 meters (NIncSimul5000_Estimado) shows a mean of 0.477 incidents with a standard deviation of 1.488. The high positive skewness (9.01) and extreme kurtosis (141.5) indicate a distribution with many low values and a few extreme outliers, making the median of 0 a more appropriate measure of central tendency. For number of simultaneous fires in same municipality (NIncSimulConcelho_Estimado), the mean is 0.921 incidents, with a standard deviation of 2.548. The skewness (8.77) and kurtosis (132.4) are both very high, suggesting that the data is heavily skewed to the right with a long tail and extreme outliers. Thus, the median of 0 incidents is a better representation. The number of simultaneous fires in same district (NIncSimulDistrito_Estimado) variable has a mean of 8.693 incidents and a standard deviation of 22.278. The skewness (9.14) and kurtosis (127.8) are very high, indicating that the data has a long right tail with significant outliers. Given this, the median of 3 incidents provides a better measure of central tendency.

For number of simultaneous fires in radius of 5000 meters in the first 90 minutes (NIncSimul500090_Estimado), the mean is 0.351 incidents, with a standard deviation of 0.972. The positive skewness (6.35) and high kurtosis (71.8) suggest that the distribution is heavily right skewed with outliers, making the median of 0 incidents more appropriate.

Number of simultaneous fires in same municipality in the first 90 minutes (NIncSimulConcelho90_Estimado) shows a mean of 0.670 incidents and a standard deviation of 1.589. The data has high skewness (6.22) and kurtosis (65.1), indicating a distribution with a long right tail and several outliers. Therefore, the median of 0.00 incidents is more suitable for representing this data. Finally, number of simultaneous fires in same district in the first 90 minutes (NIncSimulDistrito90_Estimado) has a mean of 6.227 incidents with a standard deviation of 12.560. The skewness (6.95) and kurtosis (72.9) are relatively high, pointing to a right-skewed distribution with some extreme values. The median of 2.00 incidents is a better measure of central tendency for this variable due to the presence of significant outliers.

In summary, across these variables, the high skewness and kurtosis values suggest that the median is generally a more reliable measure of central tendency than the mean, due to the presence of extreme values and right-skewed distributions.